Planned architecture

UML diagrams of the iOS app, the back end and the database, and the hosting options compared for F-11 (option C chosen). Drawn from the decisions (ADRs) and stories in `docs/`.

Version 1.6 · 12 November 2026 · own API, Postgres and S3 at Scaleway; signatures in the app, sealed by the API · kept up to date by US-004, US-006, US-050 and US-070

At a glance

Front end	One iOS app in Expo · React Native · TypeScript, built in the cloud with EAS (no Mac needed). Installed on the two staff iPhones by Ad Hoc distribution. Online only: no business data is kept on the phones.
Back end	The developer's own API (TypeScript, Fastify) at Scaleway, Paris: logins, a check on every request, 5-minute links to private files, PDF rendering. Staging on his homelab.
Database	Managed PostgreSQL: 18 business tables and seven auth tables. Status changes through SQL functions only; signed data insert-only; audit generation awaits US-007.
Signatures	A pad in the app: the client draws once, the API stamps it on every page and seals the PDF with the company's certificate and a timestamp. Optional stored Retour signature (F-28).
Contract PDF	Rendered by the API with Chromium, in the language chosen in the preview. Archived with its SHA-256.
Operations	Nightly encrypted backup to a second EU provider, Sentry EU without personal data, deployments by CI.
Accounts	Apple: the company. Expo, GitHub, Scaleway, backups, Sentry: the developer (F-02), who pays the hosting (F-07).
Still open	The spike on stamping and sealing (US-009, part 2); legal checks L-01 to L-07. F-11 decided: C (pages 14–15).

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How to read the diagrams

Deployment diagram p. 2–3
where each part runs, and who calls whom

Component diagram p. 4–5
the parts of the software, and what uses what

Class diagram p. 6–9
the tables, their fields and their links

State machine p. 10–11
the statuses, and what moves a record from one to the next

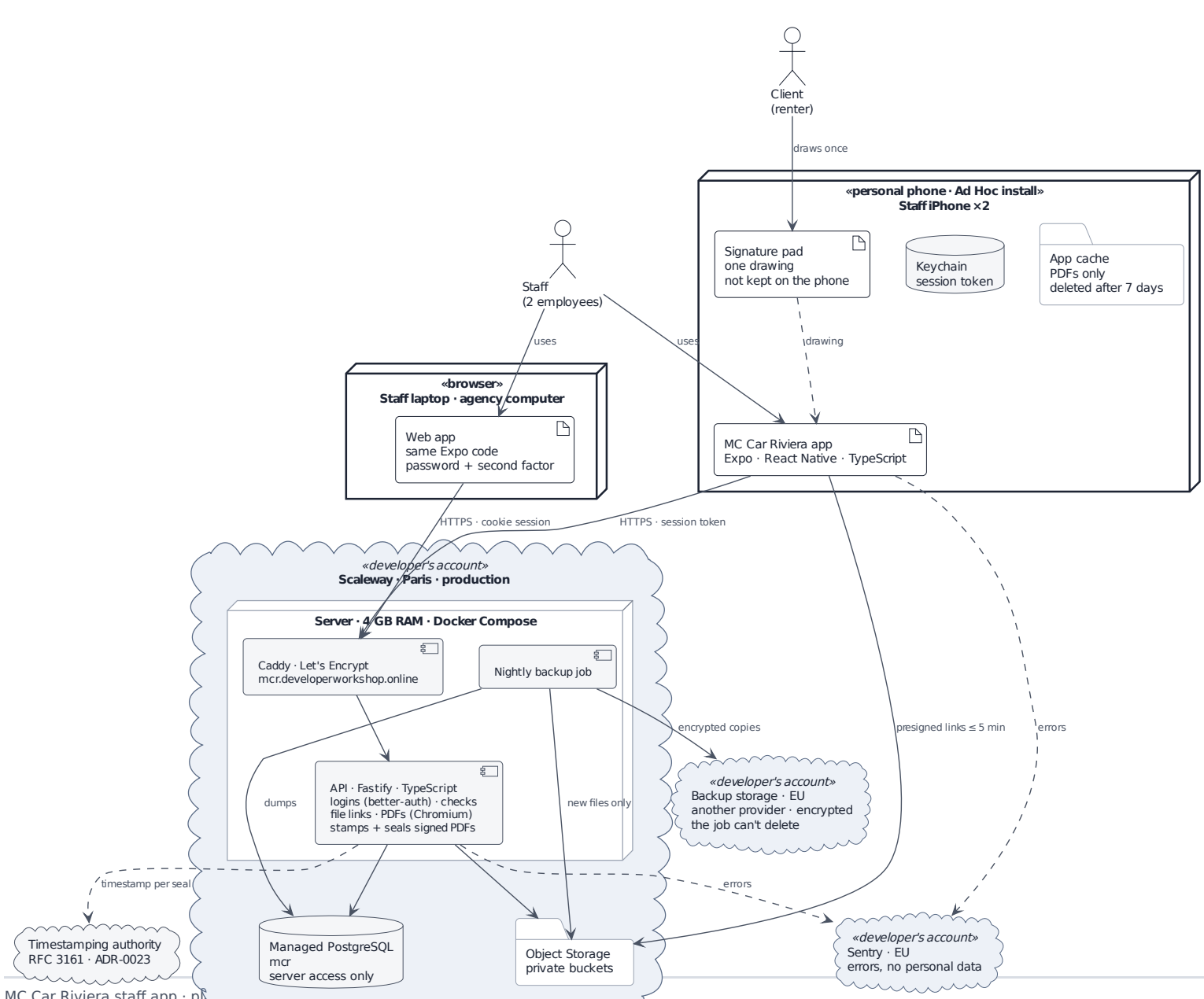
Sequence diagram p. 12–13
one workflow, message by message, from top to bottom

Colours

Company's account (Apple)
Developer's accounts (F-02)
Phone or computer

2 System overview: what runs where

UML deployment diagram · the app in daily use



Company's account (Apple)

Developer's accounts (F-02)

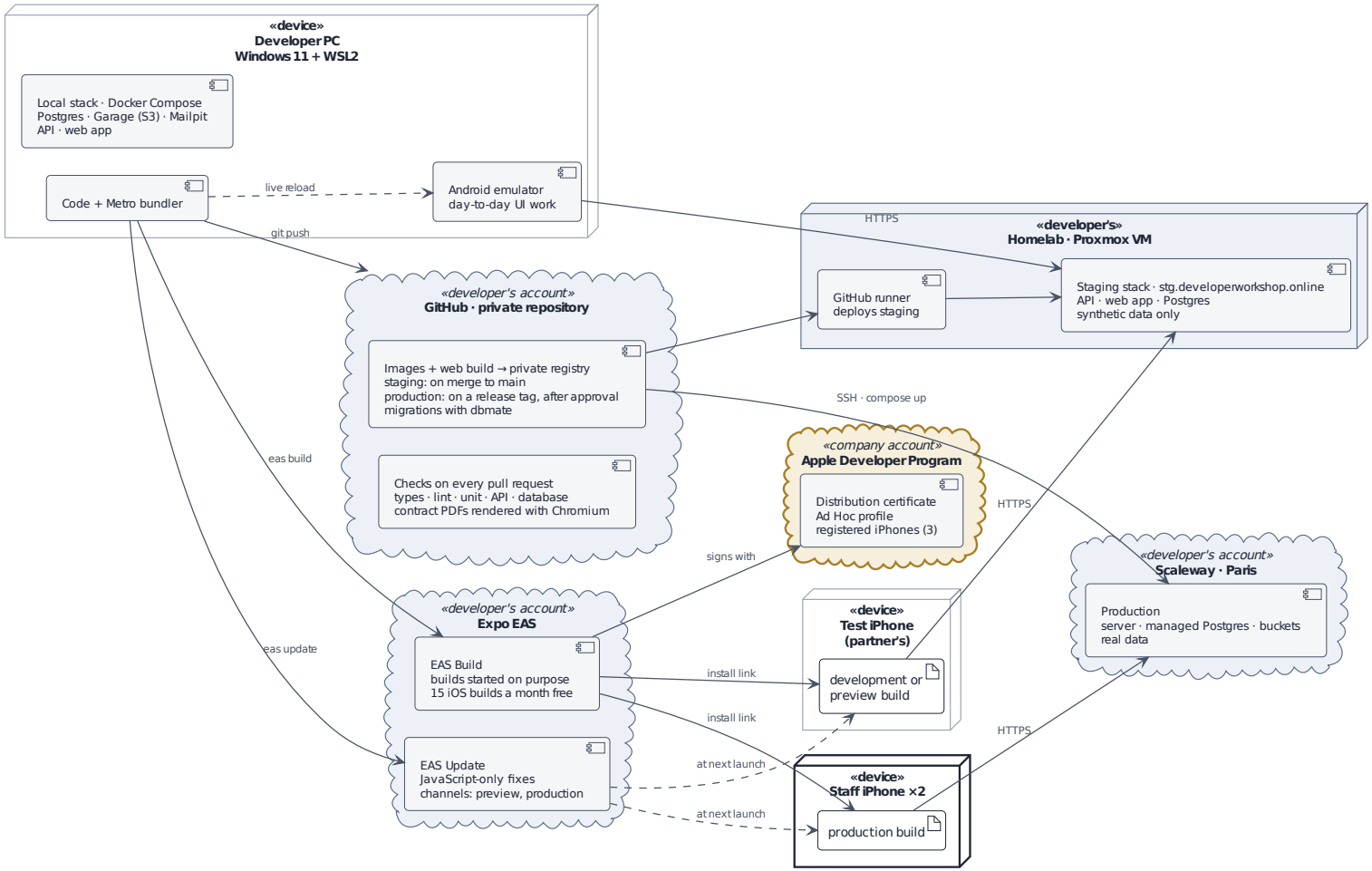
Phone or computer

Key points

- The same app runs on the iPhones and, as a web app, on the staff laptops and the agency computer (F-19). No local database and no offline mode; only PDFs to share sit in the iPhone's cache, for 7 days at most.
- Every call goes over HTTPS to the developer's own API at Scaleway, in Paris, at mcr.developerworkshop.online. The app holds only its session (Keychain on the iPhone, a protected cookie on the web), never a database password, a storage key or the signing certificate.
- The client draws once on a pad in the app. The API stamps the drawing on every page, seals the PDF with the company certificate and a timestamp, stores it, and closes the step in one transaction (ADR-0023).
- Files are private. The app reaches them through presigned links, issued by the API, that expire within 5 minutes.
- A nightly job on the server copies the database and the new files, encrypted, to a second EU provider. Its key can't delete anything there.

3 Build, release and development

UML deployment diagram · from the developer's PC to the staff phones



Company's account (Apple)

Developer's accounts (F-02)

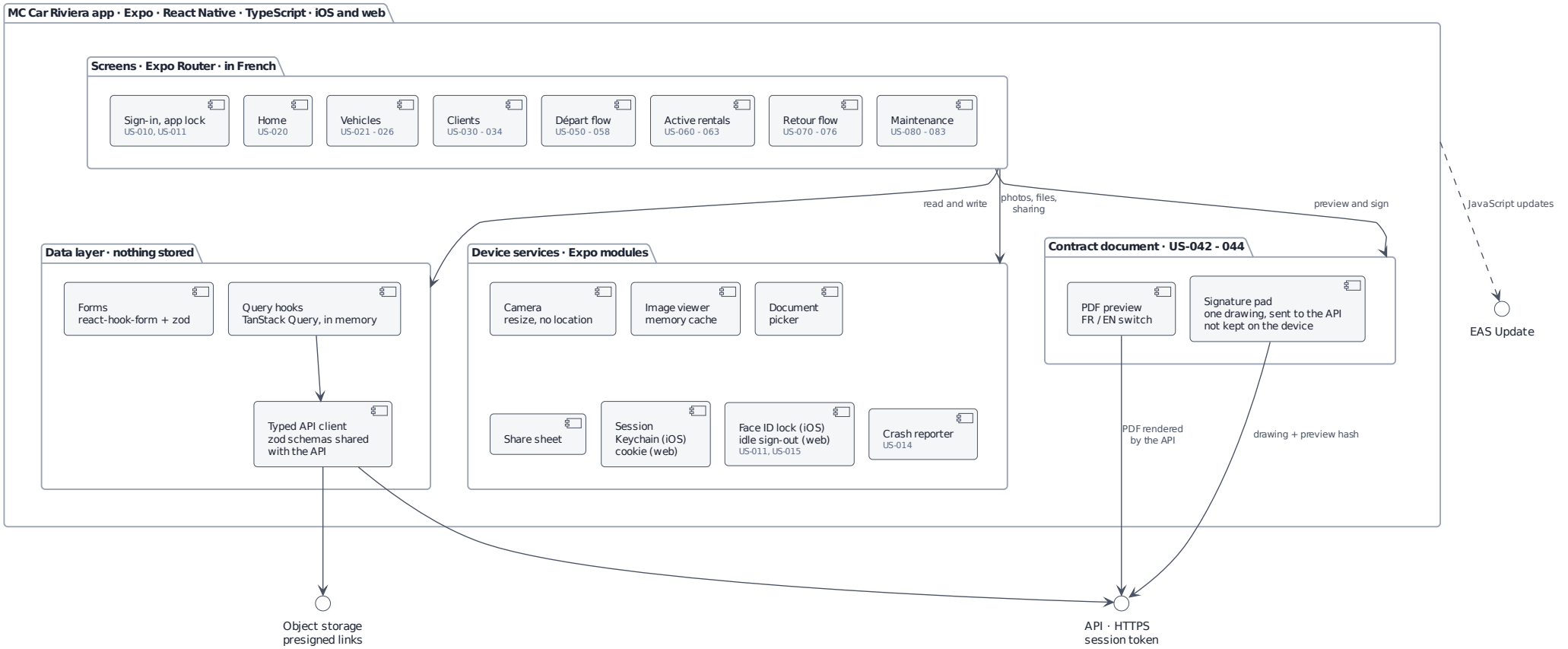
Phone or computer

Key points

- No Mac: EAS Build compiles the iOS app in the cloud. Builds are started on purpose; the free plan includes 15 iOS builds a month.
- Every build is installed by Ad Hoc internal distribution, signed with the company's Apple certificate. EAS holds it through the developer's Admin access; if the collaboration ends, the company revokes it and issues a new one.
- Staging runs on a VM on the developer's homelab, with synthetic data only. It serves the test iPhone and the Android emulator; production, at Scaleway, serves only the two staff phones.
- CI checks every pull request, including contract PDFs rendered with Chromium. Staging is deployed on merge to main; production on a release tag after manual approval. JavaScript-only fixes go out as EAS Updates.

4 Front end: the iOS app and the web app

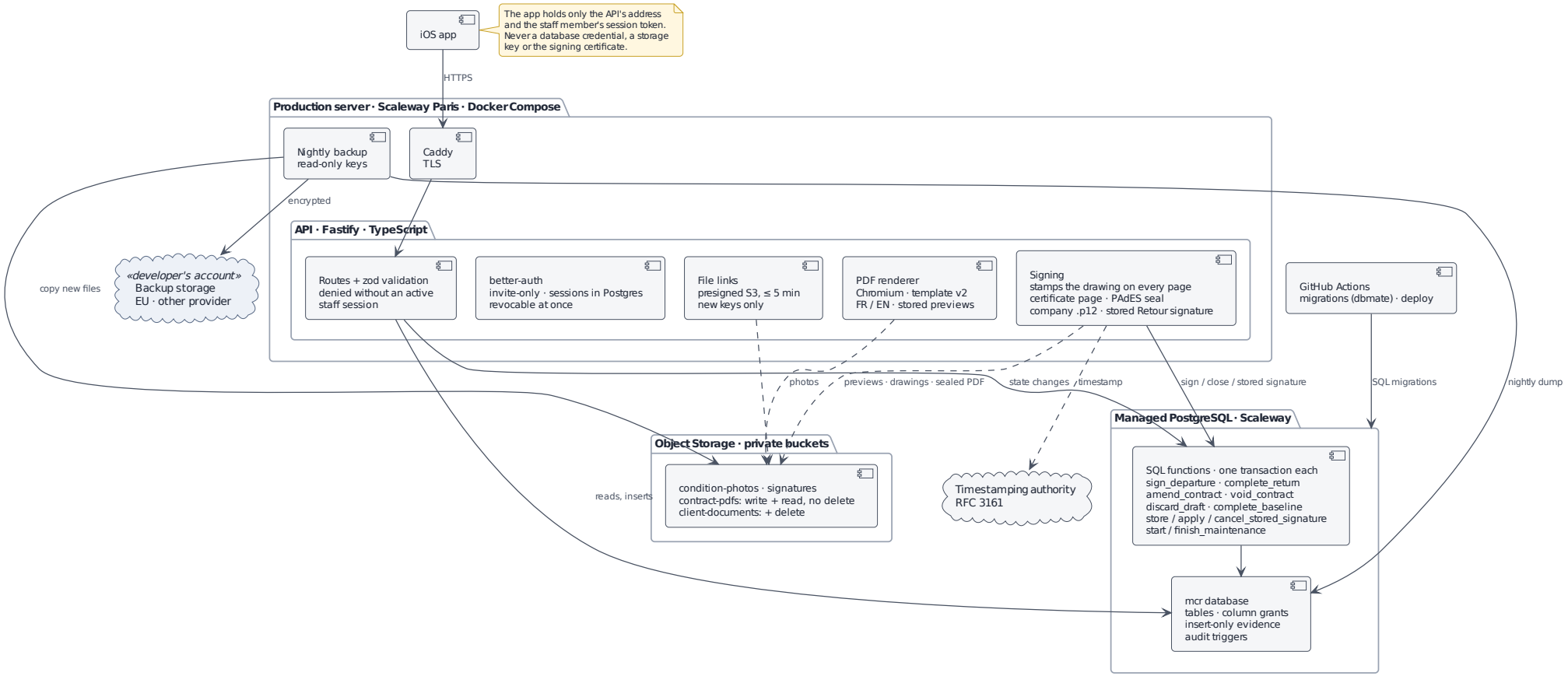
UML component diagram · one Expo codebase for iOS and the web; each screen lists the stories that build it



- One codebase for the iPhone and the web (F-19). Screens are in French, built with translation keys so English can be added later.
- Server data is kept in memory only (TanStack Query, no persister). Forms are checked with zod before anything is sent.
- The app renders no PDF: it shows the API's preview, with a French / English switch, then a full-screen signature pad. The drawing goes to the API with the preview's hash and isn't kept on the device.
- Photos are resized and stripped of their location before upload. The session token is in the Keychain. Crash reports carry no personal data.

5 Back end: the developer's API

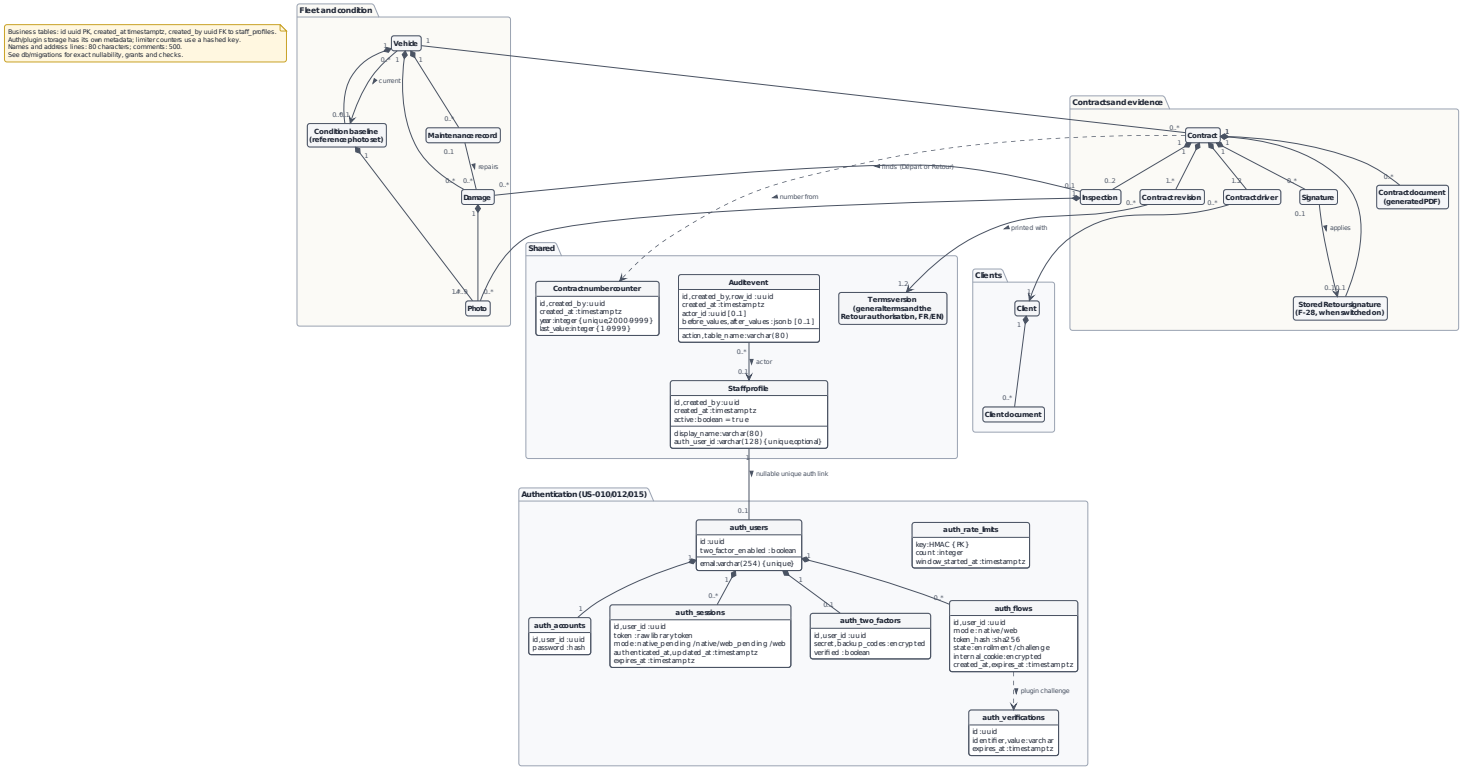
UML component diagram · Scaleway, Paris; the API checks every request, the database enforces the rules



- Every route needs an active staff session (better-auth), denied by default; a test lists every route. Revoking a session works at once.
- Every status change (sign, close, amend, void, maintenance) is one SQL function running in one transaction, so two phones can't rent the same car. Status columns are read-only for the API's role, evidence tables accept inserts only, and audit triggers record every write.
- The API stores each preview, stamps the client's drawing on every page, appends a signature certificate page and seals the file with a timestamp; it stores the sealed PDF before the contract changes state. A stored Retour signature, when the company switches it on, is applied the same way (F-28).
- The API's storage key can't delete evidence. Database credentials, storage keys and the signing certificate live only on the server and in protected CI environments.

6 Database: business and authentication

UML class diagram · overview; the fields are on the next three pages

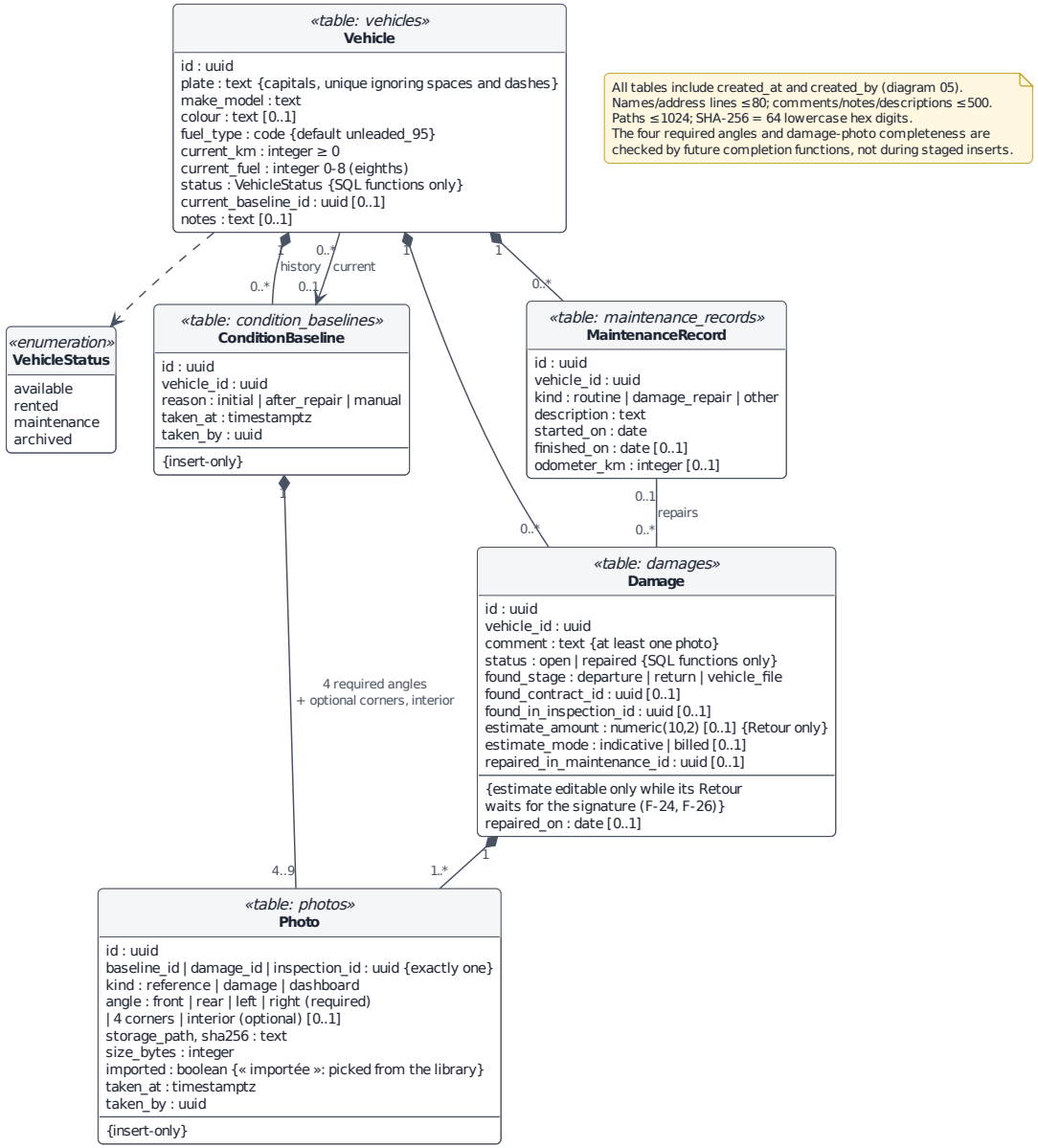


- Working table names from US-006. Once the migrations exist they are the reference, and these diagrams follow them.
- US-010/012/015 add seven auth tables and profile activation. Better Auth stores identities, hashed passwords, Postgres sessions and encrypted factors; pending native/web flows cannot access business routes; both modes require MFA, with native sessions capped at 90 days. Auth records use library metadata; business creation attribution stays unchanged.
- A contract is a header plus numbered revisions. Revision 1 is the Départ as signed; each amendment adds a revision with a full snapshot, recorded in the app and not printed (F-22).
- Condition evidence is reused: a Départ's inspection points to the vehicle's current reference photo set and to the damage open at that moment (drawn on « What each handover records »).

Notation ◆— part of (belongs to one owner) —› refers to **1 · 0..1 · 0..*** how many **[0..1]** optional field **{…}** rule

7 Database detail: fleet, photos, damage, maintenance

UML class diagram · vehicles, condition_baselines, photos, damages, maintenance_records



Key points

- A reference set needs front, rear, left and right (F-23); the four corners and the interior are optional. It is taken with the in-app camera.
- A damage can be found at a Départ (never charged to that renter, F-25) or at a Retour, where it can get an estimate, indicative or billed (F-24).
- The optional dashboard photo (km and fuel) belongs to a handover's inspection, not to the reference set: reused later, it would show an old odometer reading.
- Finishing a damage repair closes the repaired damage. The car can be rented again only once a new reference set exists.
- Photos and reference sets are insert-only. Statuses change only through SQL functions.

Notation

◆— part of (belongs to one owner)

—› refers to

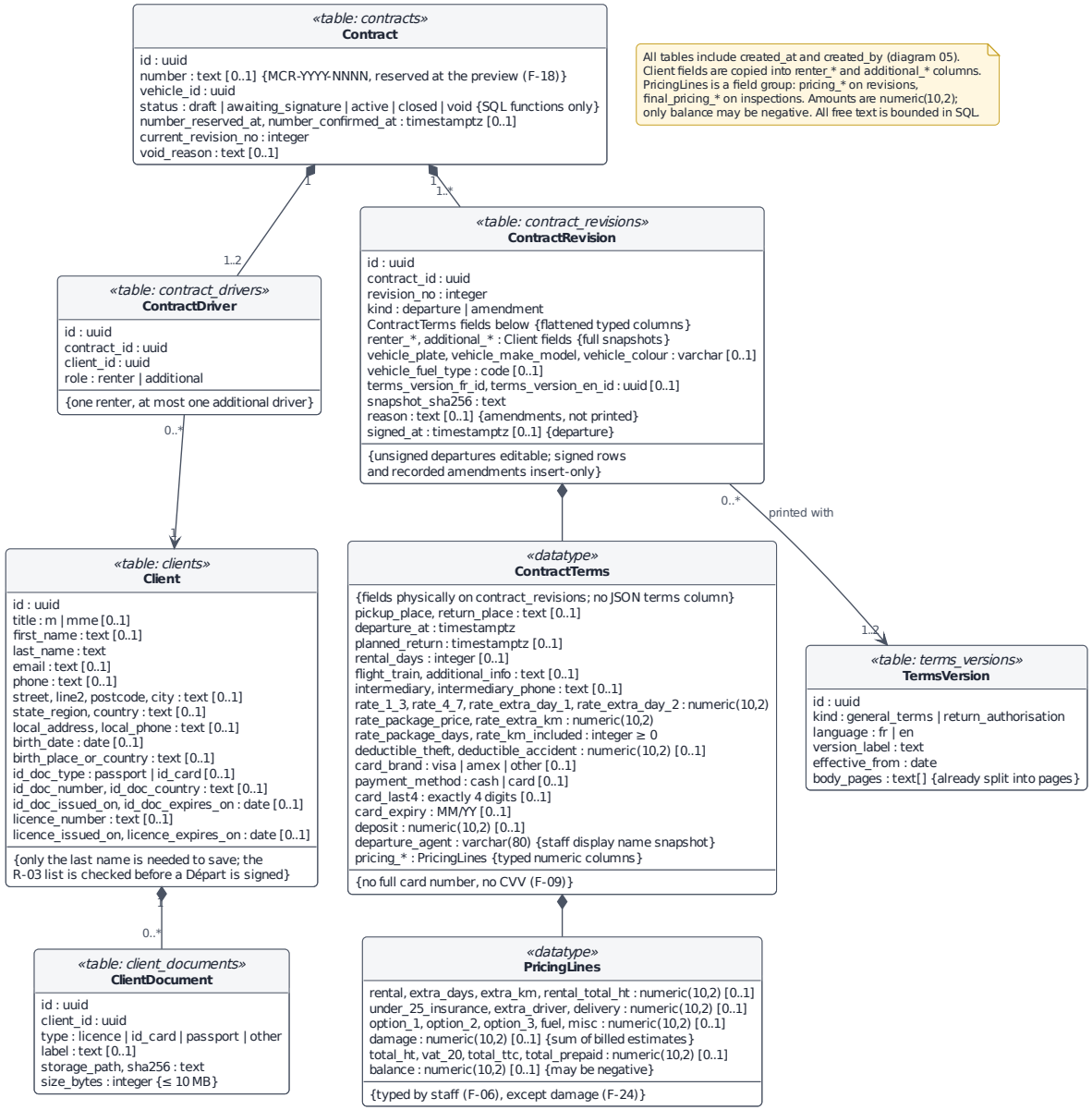
1 · 0..1 · 0..* how many

[0..1] optional field

{…} rule

8 Database detail: clients, contracts and their terms

UML class diagram · clients, client_documents, contracts, contract_drivers, contract_revisions, terms_versions



Key points

- Signing freezes revision 1: the client and vehicle values are copied as printed, so later edits to a client or a vehicle never change a signed contract.
- Card: brand, last four digits and expiry only. No column can hold a full card number or a CVV (F-09, PCI DSS).
- Every amount is typed by staff (F-06), except the « Dommages » line, the sum of the billed estimates (F-24). The app also computes km driven and extra km.
- Each revision records the French and English terms versions in force when it was signed.

Notation

◆— part of (belongs to one owner)

—› refers to

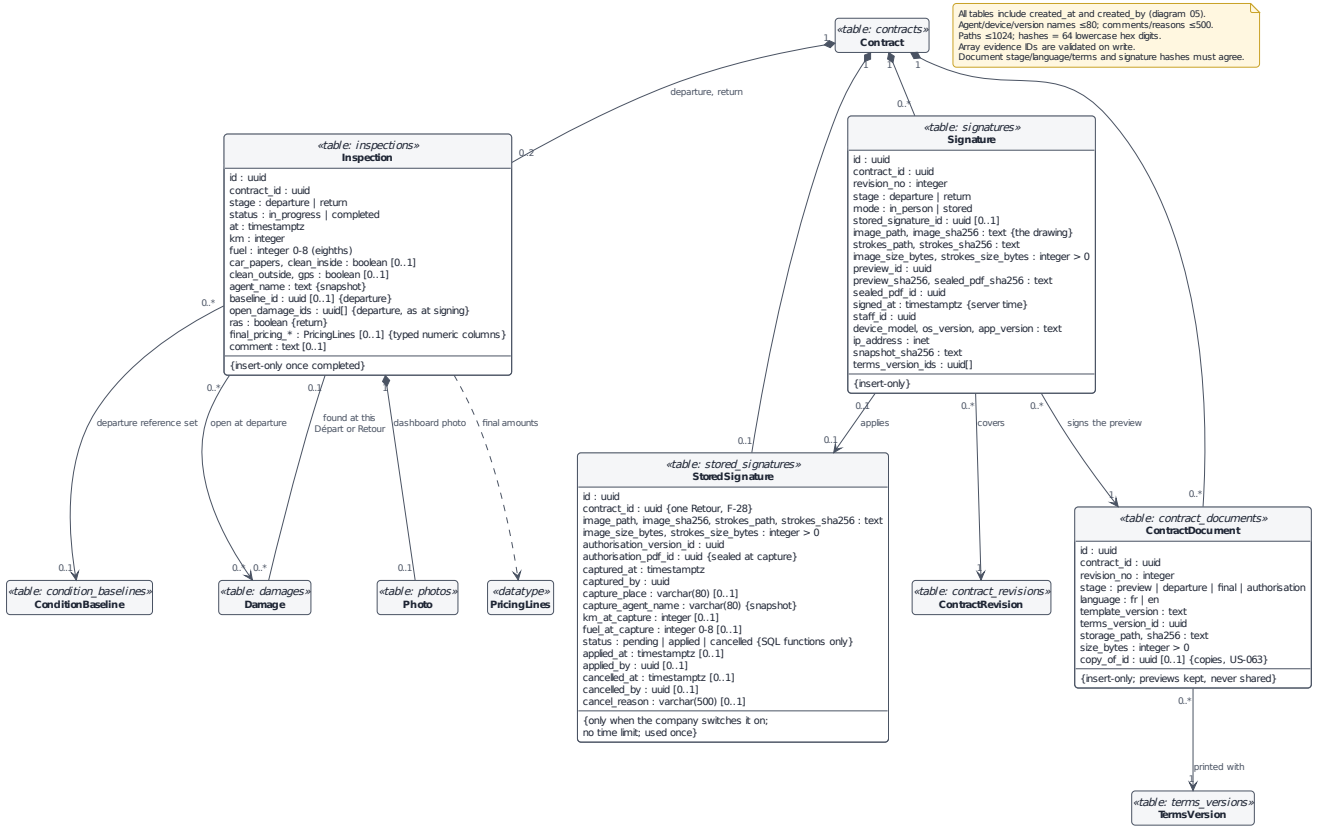
1 · 0..1 · 0..* how many

[0..1] optional field

{…} rule

9 Database detail: what each handover records

UML class diagram · inspections, signatures, stored_signatures, contract_documents (grey: drawn in full on earlier pages)

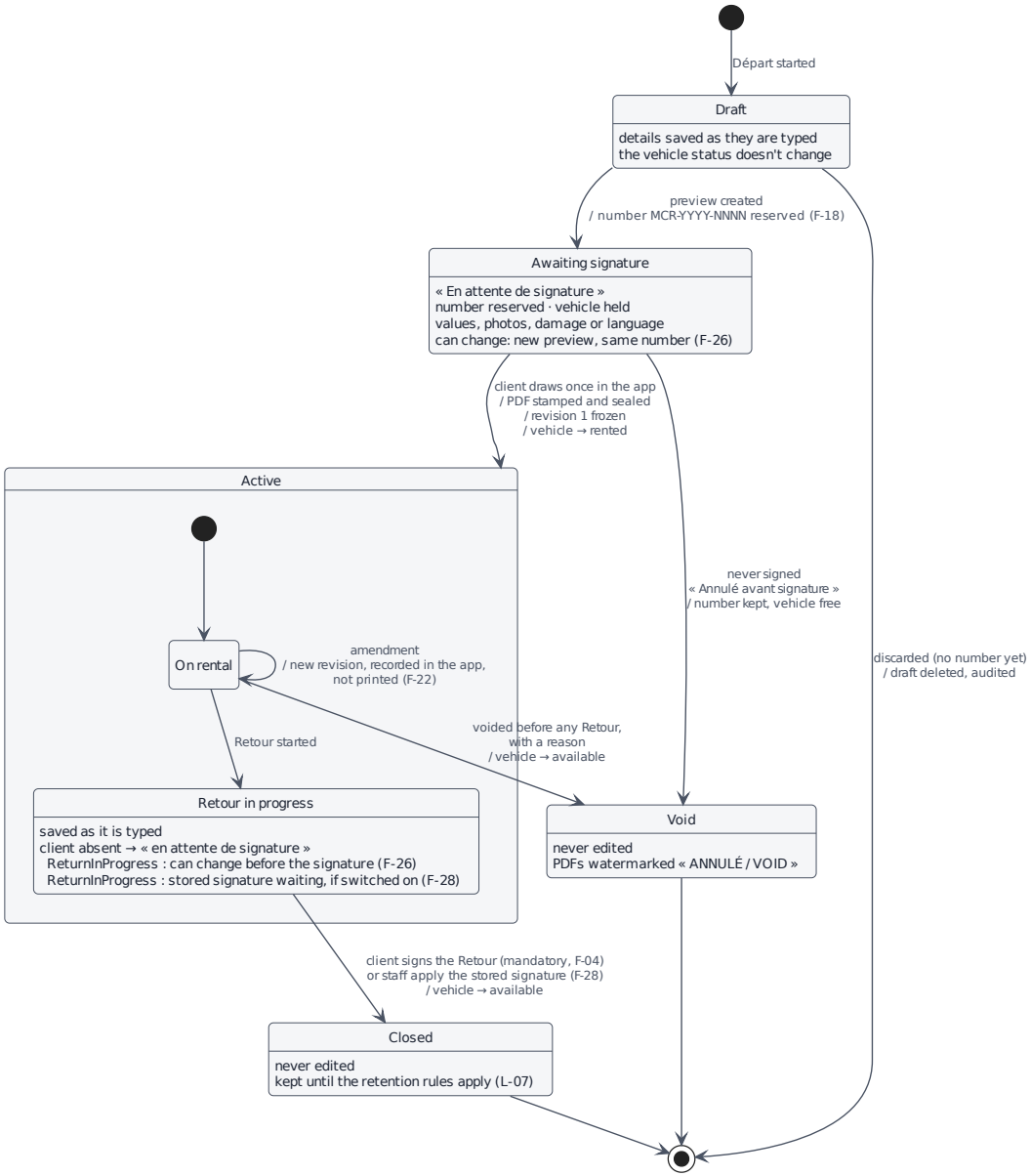


- Each signature has an evidence record: the drawing, the preview it signed and the sealed PDF it is in, with their hashes, server time, staff member, device, versions, IP address, signing mode, and the hash of the data that was signed. The departure drawing is reused on the final PDF (F-17).
- A stored Retour signature (F-28) is pending until staff apply it once to that Retour, or cancelled; its status changes only through SQL functions.
- Every PDF is archived with its SHA-256, its template version and its terms version. Copies (US-063) point to the sealed original.
- Completed inspections, signatures and PDF records are insert-only. Until the client signs, the Retour's km, fuel, amounts and comments can still be corrected; whether a photo or a damage entry can be taken back is open (R-10 Q5).

Notation ◆— part of (belongs to one owner) —› refers to **1 · 0..1 · 0..*** how many **[0..1]** optional field **{…}** rule

10 Lifecycle of a contract

UML state machine · every transition is an SQL function called by the API, recorded in the audit trail

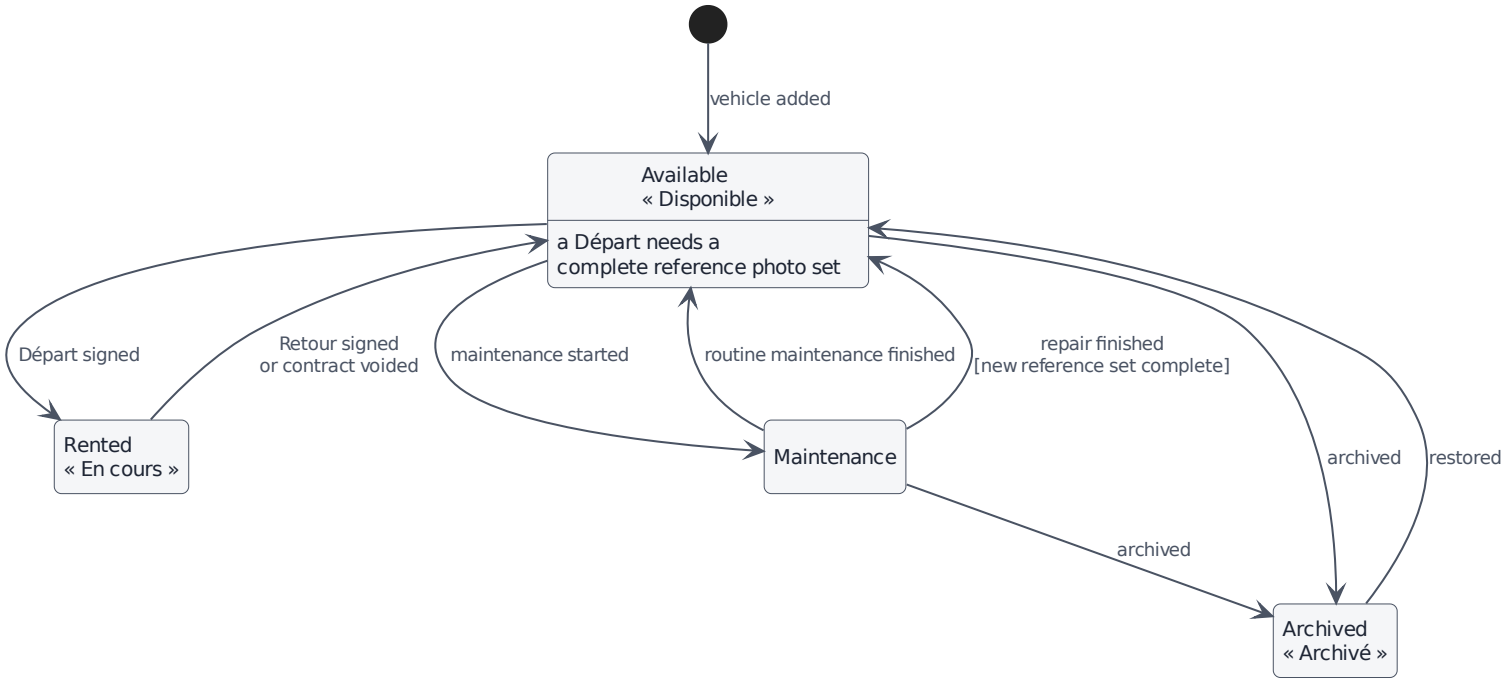


Key points

- A draft never changes the vehicle's status. Before the preview it can be discarded; once the preview reserves a number, it can only be cancelled, and is kept as « Annulé avant signature » (F-18).
- Signing the Départ confirms the number, freezes revision 1 and marks the car rented, in one transaction.
- An amendment adds a revision, recorded in the app; it has no page in the PDF and isn't signed in the app (F-22). Until the client signs, a Départ or a Retour can take new photos and damage (F-26).
- A contract can be voided, with a reason, only before its Retour. The app never deletes a signed contract.

11 Lifecycle of a vehicle

UML state machine · the labels in « » are what staff see in the app

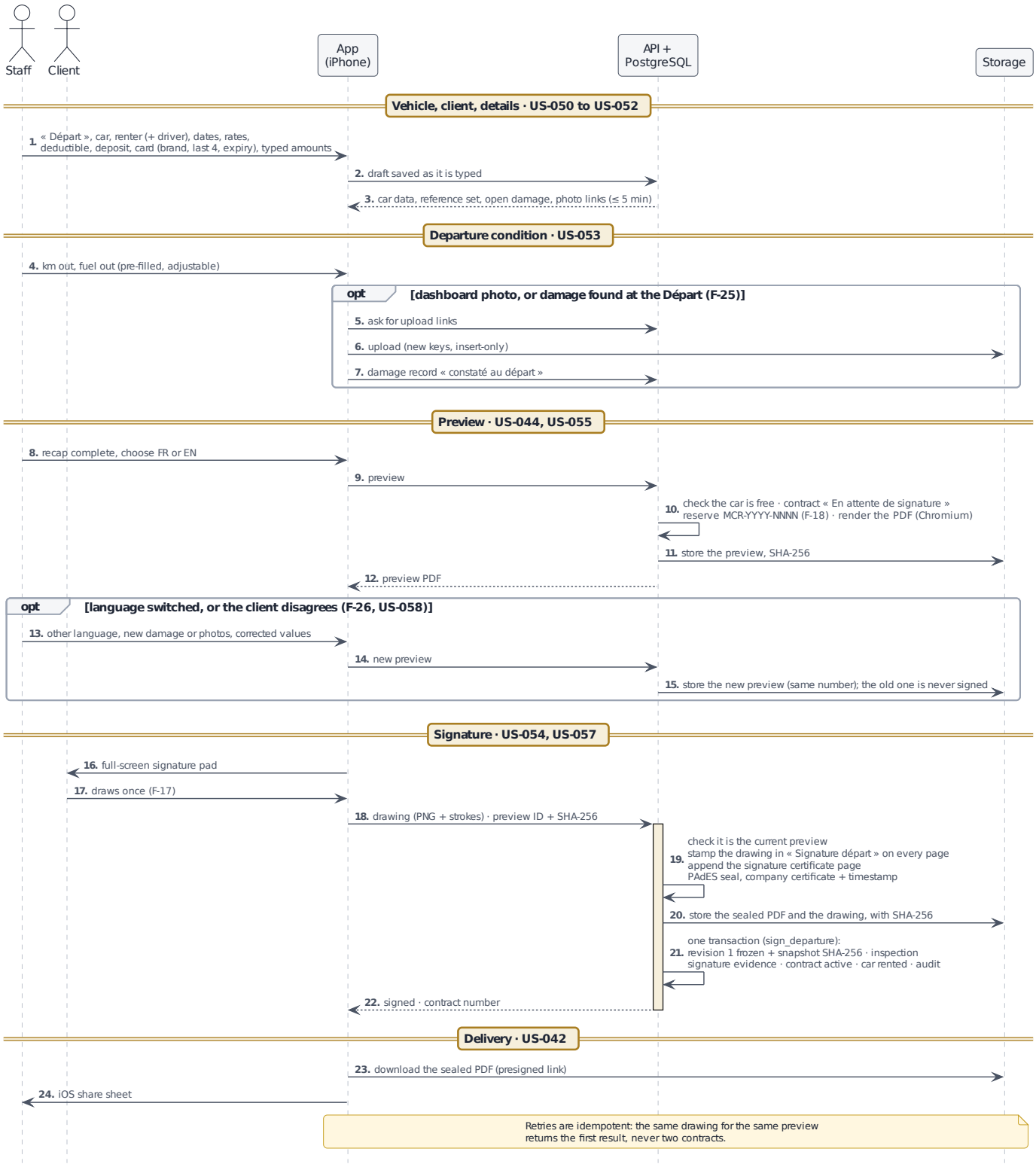


Key points

- A Départ is offered only for available cars with a complete reference photo set.
- Routine maintenance needs no new photos. After a damage repair, a new complete reference set is required before the car is available again.
- Only an available car, or one in maintenance, can be archived. « Restaurer » brings it back; nothing is deleted.
- Statuses change only through SQL functions, in the same transaction as the contract or maintenance change.

12 Départ: from choosing the car to the sealed PDF

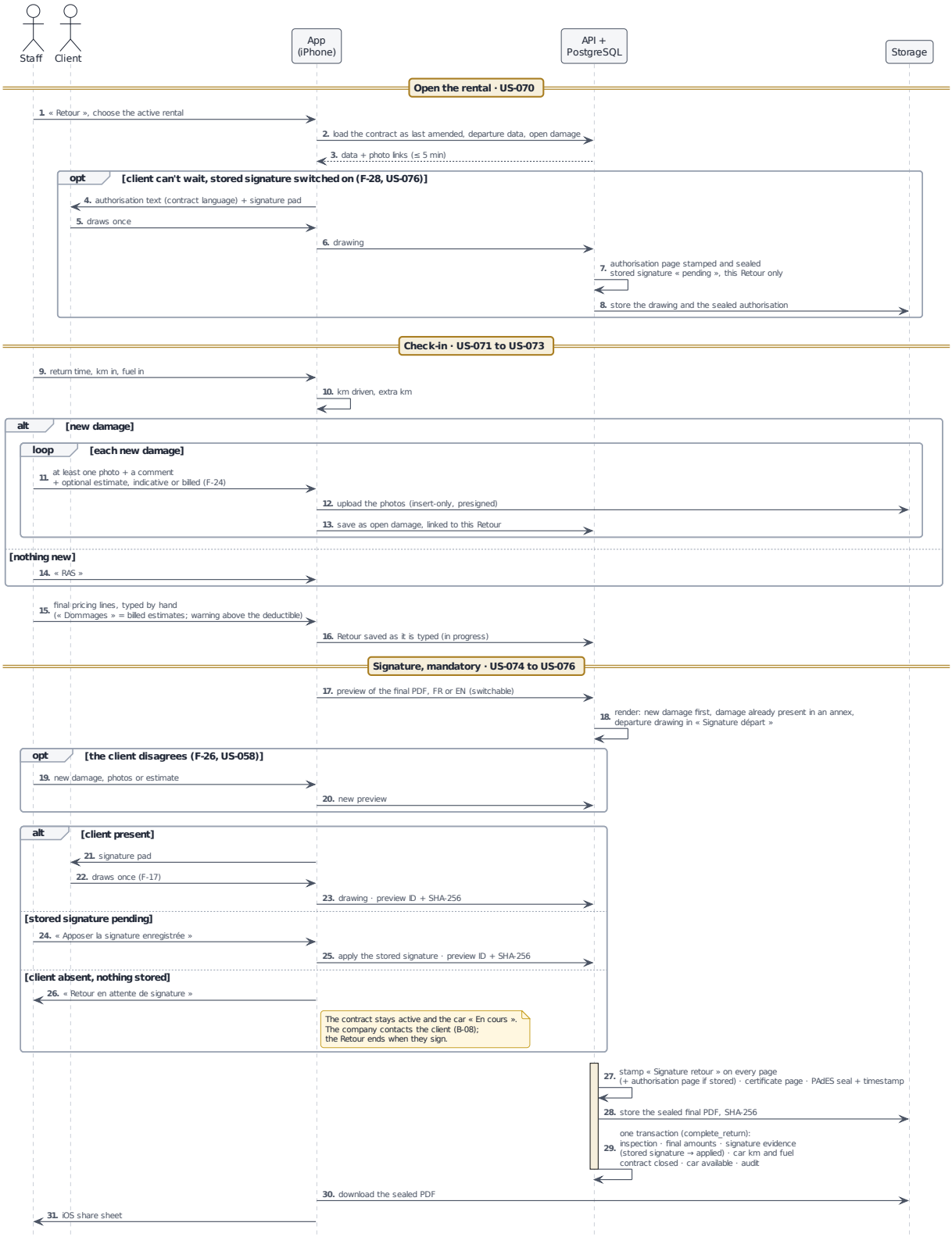
UML sequence diagram · US-050 to US-058, US-044



- The language is chosen in the preview, and can be switched until the client signs (F-17). The PDF carries its contract number already, so the number is reserved at the preview (F-18).
- The client draws once on the pad; the API stamps the drawing in the signature box of every page of the preview they saw, and seals the PDF with the company's certificate and a timestamp. A change before the signature makes a new preview (F-26).
- The contract becomes active only once the sealed PDF is stored, in one transaction. If the car was taken meanwhile, nothing is signed in the app.

13 Retour: check-in, mandatory signature, final PDF

UML sequence diagram · US-070 to US-076, US-044



- The return signature is mandatory (F-04). Without the client, the Retour waits « en attente de signature » and the car stays « En cours ».
- Km driven and extra km are the only values the app computes. The final amounts are typed (F-06).
- In the final PDF the client's drawing fills the « Signature retour » box of every page, or the stored signature does when the company switched it on and the client recorded it at the drop-off (F-28). The « Signature départ » boxes show the departure drawing (F-17). New damage comes first (F-24).

14 Why an API between the phone and Postgres (F-11)

Your question of 9 October, answered on 3 November with option C · the four options side by side

Traffic isn't what costs money

Two staff members and a few dozen requests per handover are nothing for any of these options. What costs money is storing the files, and the security layer someone has to build and run.

The phone can't talk to Postgres directly

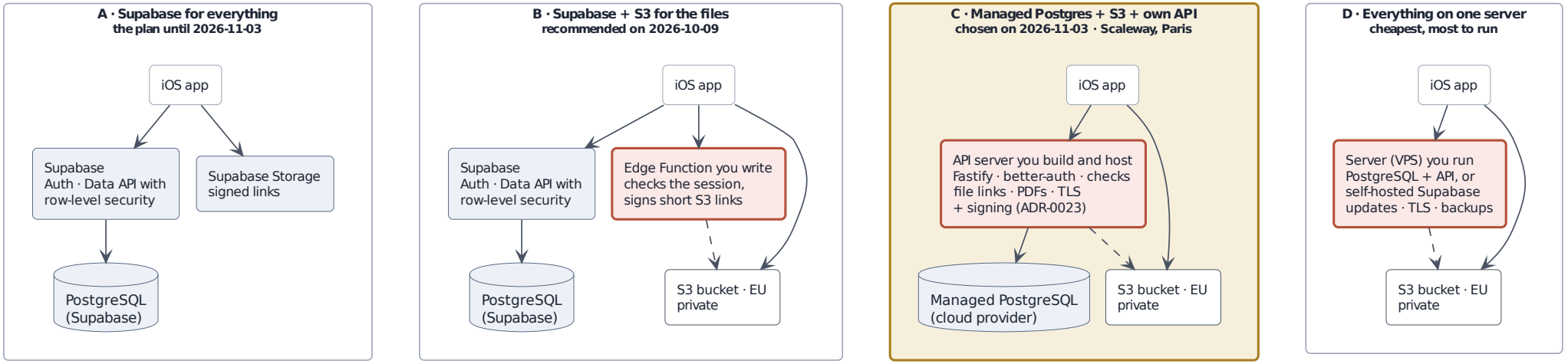
It would need a database password on two personal phones, and Postgres open to the internet. Something must sit in between: sign staff in, check every request against the access rules, and hand out short-lived links to private files. With option C, that layer is the developer's own API.

What C costs

About €25–30 a month: the managed Postgres (about €11), a server with at least 4 GB of RAM for the API and Documenso (about €15), cents for the files. And weeks of extra work: logins, permissions, file links, deployment, monitoring.

What C gives

No free-plan pause or quota, the data on infrastructure the developer controls, in France, one language from the app to the API, and room for Documenso next to the API.



Run by a provider · Built or run by the developer · S3-compatible bucket

15 Hosting options compared (F-11: C chosen)

Approximate list prices before VAT, checked on 9 October 2026 · every option stays portable (Postgres dump, S3-compatible files)

	A · Supabase for everything the plan until 3 November	**B · Supabase + S3 for the files** recommended on 9 October	**C · Managed Postgres + S3 + own API** chosen on 3 November	**D · Everything on one server** cheapest, most to run
What runs where	Logins, database, API and files on Supabase (EU)	Logins, database and API on Supabase Free; files in an EU S3-compatible bucket (e.g. Scaleway, Paris)	Managed Postgres, a small server for your own API, and an S3 bucket	One small server (e.g. Hetzner CX23, €5.49) with Postgres and an API, or self-hosted Supabase; files in S3
Per month, under 1 GB of files	€0	€0 + cents	≈ €17; ≈ €25–30 with Documenso on a 4 GB server	≈ €6–8
Per month after ≈ 200 rentals	$25 (Pro)	Cents: the text data stays under 500 MB for years	The same, plus cents for files	≈ €6–8
Extra work for you	None	1–2 days: one function for file links, and the bucket set-up	Weeks: logins, sessions, permissions, the API itself, TLS, monitoring	Days to set up, then updates, security, backups and monitoring of a server holding ID documents
Who patches and watches it	Supabase	Supabase and the bucket provider	The provider for the database; you for the API server	You
Limits to know	Free plan: no service guarantee; pauses after a week without activity	Free-plan limits for the database, as in A	None of the free-plan limits	Your one server is a single point of failure

Decision (3 November): C

Your own API server in TypeScript with Fastify (F-12), a managed Postgres and S3 storage, at Scaleway in Paris (F-14), with staging on your homelab. Signatures with a self-hosted Documenso next to the API, replaced on 10 November by signing in the app, sealed by the API (ADR-0023). Recorded in ADR-0019; Supabase (ADR-0004) is superseded.

What C adds since 9 October

- A server with 4 GB of RAM to start, checked after the spike: about €25–30 a month in all.
- An e-mail provider (F-15), first chosen for Documenso, kept for later e-mail.
- PDF rendering on the server with Chromium (F-13), and the nightly backup run from the server.

Still to watch

- One server: while it is down, no handover can be signed. Uptime alerts and the paper fallback cover it.
- Scaleway becomes a sub-processor in your data processing agreement with the company (L-03).

Prices: Supabase pricing page (Free and Pro plans) · Scaleway price lists for Object Storage (€0.01606 per GB a month, Paris, multi-AZ) and Managed Databases (DB-DEV-S, €0.0156 an hour) · Hetzner's cloud price change of 15 June 2026 (CX23 from €3.99 to €5.49 a month). Check them again before committing.